

Akula Chandu Swamy

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SUMMARY

Full Stack Python Developer with 4+ years of experience in designing scalable backend systems and dynamic frontends for finance and healthcare domains. Proficient in Python, FastAPI, Django, ReactJS, and containerized microservices using Docker and Kubernetes on AWS. Skilled in developing RESTful APIs, implementing CI/CD pipelines, and integrating caching and messaging systems like Redis, Celery, and RabbitMQ. Strong experience with data engineering workflows, API security (OAuth2, JWT), and HIPAA-compliant cloud solutions. Adept in Agile environments with a solid understanding of SDLC, cloud platforms (AWS, Azure, GCP), and DevOps best practices.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Languages & Frameworks: Python, Django, Flask, FastAPI, JavaScript (Node.js), HTML5, CSS3, Bash

API & Backend Development: RESTful APIs, GraphQL, OpenAPI/Swagger, gRPC, Celery, Redis, RabbitMQ

Web Technologies & Front-End Integration: Jinja2, React (basic), Bootstrap, AJAX, JSON, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Cassandra

Data Engineering & Pipelines: Pandas, NumPy, PySpark, Airflow, DBT, ETL/ELT, Kafka

DevOps & MLOps: Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, MLflow, DVC

Cloud Platforms: AWS (Lambda, S3, EC2, RDS, CloudWatch), Azure, GCP

Testing & Code Quality: PyTest, UnitTest, Postman, Selenium (Basic), SonarQube, Black, Flake8

Tools & Platforms: Git, GitHub, Bitbucket, Jira, Confluence, VS Code, Linux

Security & Compliance: OAuth2, JWT, HIPAA (healthcare), PCI DSS (finance), OWASP Top 10

Operating System: Windows, Linux, Mac

EXPERIENCE

Capital One, USA | Full Stack Python Developer

Nov 2023 - Current

- Designed and developed RESTful APIs using Python, FastAPI, and PostgreSQL, enabling secure, high-speed financial data processing with reduced API latency by 30% for real-time banking applications.
- Built interactive and responsive web interfaces using ReactJS, Redux, JavaScript and Bootstrap, improving the digital customer experience and reducing front-end bugs by 25% across multiple user journeys.
- Implemented containerized microservices with Docker and Kubernetes (EKS) on AWS, ensuring highly available deployments, automated scaling, and consistent environments across Dev, QA, and Prod.
- Developed CI/CD pipelines using Jenkins, GitHub Actions, and Terraform, enabling infrastructure as code (IaC) and automated testing, which accelerated deployment cycles by 40% and reduced manual errors.
- Integrated Redis for caching and Celery with RabbitMQ for asynchronous task queues, optimizing system performance and throughput during peak transaction loads in Capital One's loan processing systems.
- Worked extensively with AWS services (EC2, EKS, RDS, S3, Lambda, DynamoDB, CloudWatch) to build scalable cloud-native applications with enhanced monitoring, logging, and cost optimization strategies.

Rlogical Techsoft Pvt. Ltd, India | Python Developer

Jan 2019 - Dec 2021

- Engineered backend services using Python and FastAPI to support a scalable telehealth platform, allowing patients to schedule, join, and review virtual consultations across web and mobile apps.
- Integrated real-time vitals data streaming using WebSockets and MQTT, enabling continuous remote monitoring of heart rate, SpO2, temperature, and ECG from IoT medical devices.
- Developed AI-driven symptom checker using Scikit-learn and NLP libraries, allowing patients to input symptoms and receive triage recommendations with over 85% accuracy.
- Created asynchronous task queues using Celery and Redis to manage video session logs, prescription generation, and follow-up reminders, reducing system latency during high-traffic hours.
- Deployed microservices on AWS using Docker, ECS, and RDS, ensuring high availability and compliance with healthcare data security standards (ISO 27001, HIPAA-aligned practices).
- Built admin dashboards and patient portals using ReactJS and REST APIs, improving user adoption and enabling doctors to securely access patient vitals, chat logs, and e-prescriptions.
- Integrated third-party APIs for SMS notifications, e-prescription generation, and video conferencing streamlining communication between patients and healthcare providers.
- Collaborated with data science and QA teams in Agile sprints using Jira, contributing to a 30% reduction in teleconsultation no-shows through data-driven appointment reminders and insights.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate.

EDUCATION

Master's in Information Studies | Trine University, Angola, Indiana

Bachelor's in Computer Science Engineering | GITAM University, India